Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Aerothermal Response of Ceramic Matrix Composites Thermal Protection System*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	*Measurement Sensors and Component*	*Aerospace Systems*	*Other Aerospace System Technologies*
Concept of Proposed Technology	The proposed research integrates advanced SiC/SiC ceramic-matrix composites, high-enthalpy plasma experiments, and novel heat flux sensors to create a robust framework for predicting and controlling material behavior in extreme aerothermal environments. The project seeks to develop and validate fast-response heat flux gauges for high temperature flows and deploy them to characterize the environment of large-scale plasma wind tunnel. The sensors, along with laser-based diagnostics will enable quantification of surface catalytic recombination effects in reactive flows and to study the response of SiC -based composites, a promising technology for thermal protection in hypersonic flight.			

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Rationale for Joint R&D Project	The Korean industry has unique capabilities to develop material and sensors for extreme environments. The lack of scalable processes for SiC/SiC ceramic matrix composites has been an obstacle to fielding these materials in hypersonic applications. In addition, the current state-of-the-art in catalytic heat flux sensors has prevented reliable assessment of material aerothermal performance. As part of this joint R&D we seek cross-pollination of technology to develop a validated aerothermal response model for high temperature ceramic matrix composites to be used in hypersonic flight.
Keywords	1. Thermal Protection System 2. SiC/SiC Composites 3. Heat Flux Sensors 4. High-enthalpy Plasma Testing 5. Extreme Environments

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)

R&D Objectives

The Center for Hypersonics and Entry System Studies at the University of Illinois Urbana-Champaign is a national epicenter for high temperature material research. The center develops and operates the Plasmatron X facility, a national core wind tunnel for thermal protection system performance evaluation under conditions that replicate the hypersonic flight environment. A principal mission of our research at Plasmatron X is to identify and quantify the mechanisms governing the aerothermal response of high-temperature composites exposed to reactive, oxygen-rich, high-enthalpy flows (1-4). By advancing the fundamental understanding of gas–surface interactions under these conditions, we aim to enable safer and more robust design of thermal protection system (TPS) materials for future hypersonic systems. To this end, we combine the unique capabilities of the Plasmatron X environment, with state-of-the-art diagnostics for reactive flows and high temperature surfaces, and the unique array of material characterization capabilities of the Material Research Laboratory at Illinois. We have established collaboration with Korean academia in this area, through strong links with the Korea Advanced Institute of Science & Technology (KAIST) in Daejeon, which we plan to extend through this effort to the Korean industry involved in the development of extreme environment technology.

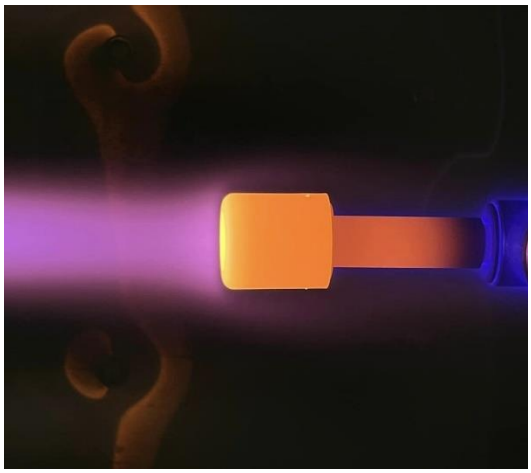

Figure 1. TPS Material testing in the Illinois Plasmatron X wind tunnel.

Specifically, this project aims to develop a comprehensive aerothermal response database for ceramic matrix composites (CMCs) under hypersonic-relevant conditions. The database will be built upon:

1. aerothermal experiments with new laser-based diagnostics conducted at the Plasmatron X facility,
2. the integration of next-generation Korean-manufactured CMCs, and
3. the deployment of advanced high-temperature sensor technologies for in-situ characterization.

We focus on silicon-carbide fiber-reinforced silicon-carbide composites (SiC/SiC) as promising candidate materials for thermal protection systems of hypersonic flight system due to their exceptional combination of high-temperature capability, structural integrity, and oxidation resistance. These composites maintain strength and stiffness well above 1,200-1,400 °C, enabling them to survive the extreme aerodynamic heating environments

	characteristic of sustained hypersonic flight. Compared to monolithic ceramics, SiC/SiC CMCs offer markedly improved toughness and damage tolerance, making them less prone to catastrophic fracture under thermal gradients or mechanical loading. Their relatively low density provides significant mass savings over traditional metallic TPS solutions. Furthermore, engineered oxide or environmental barrier coatings can enhance durability by mitigating recession in high-enthalpy, oxygen-rich flows. Herein, we aim to investigate high temperature passive-to-active oxidation and surface catalysis of SiC/SiC composites, by leveraging Korean-manufactured CMCs and sensors that will enable improved characterization of the Plasmatron X facility environment. The new measurements of oxidation and catalysis effects sought within the project are key to uncovering high temperature gas-surface interaction mechanisms for the response of CMCs to hypersonic flow.
R&D Target Specifications	• Develop gauges for high-enthalpy plasma wind tunnels for heat fluxes up to 1000 W/cm2 with uncertainties below ±10% • Develop SiC/SiC CMCs with porosities <10% • Quantify catalytic recombination efficiency for SiC-based composites at temperature up to 1600 °C • Measure active-passive transition for SiC-based composites as a function of pressure (1-10 kPa) and temperature (1000-2000 °C)
R&D Plan	We propose investigating the aerothermal performance of SiC/SiC materials for hypersonic applications. Our research plan is organized into four tasks that are described below. ***1) Task 1. Characterize the Plasmatron X hypersonic environment*** Task 1 establishes a characterization of the Plasmatron X aerothermal environment to obtain key flow quantities needed to interpret SiC/SiC oxidation and catalytic heating in later tasks. Rather than a full facility mapping, this task will generate a high-fidelity dataset describing the specific thermal and chemical conditions relevant to SiC/SiC gas–surface interactions. The characterization will focus on stagnation-point heat flux, stagnation pressure, and local gas composition and dissociation fractions (O, O_2, N_2, NO) of the plasma jet. We will leverage existing Plasmatron X diagnostics, including copper slug calorimeter-based measurements for stagnation-point heat flux, water-cooled Pitot probe for stagnation pressure, optical emission spectroscopy (OES) to reconstruct flow temperature and high-speed imaging of the jet (5-8). A key element of this task is the **integration and evaluation of Korean-manufactured heat flux sensors for high-enthalpy environments**. These include **coaxial thermocouples (TCs)** and **thin-film heat-flux gauges**. These are critical to characterize catalytic recombination effects in reactive flows. O-atom catalytic recombination at the surface is a first-order contributor to the total heat flux in high-enthalpy facilities, often comparable to or exceeding purely convective heating; quantifying this contribution is therefore essential to correctly interpret TPS performance under Plasmatron X conditions. Coaxial TCs and thin-film heat-flux gauges offers a highly responsive, robust approach for measuring surface heating in transient aerodynamic environments. An ideal coax-TC design consists of a type-T thermocouple formed by swaging a copper tube around a centrally aligned constantan pin, separated by a thin insulating layer (e.g., MgO). A copper coating deposited on the windward tip creates the thermoelectric junction directly at the surface, enabling near-instantaneous registration of temperature changes as external heat flux varies. When installed flush with the outer mold line of a copper calorimeter body, the sensor's early-time temperature history – prior to the arrival of the internal heat wave – can be analyzed using a 1D

inverse-conduction method analogous to a null-point calorimeter. This enables accurate, fast-response heat-flux measurements during rapid spatial sweeps through heated flows. Compared to traditional slug calorimeters, coax TCs provide near-zero thermal inertia at the junction, high temporal resolution, and the ability to resolve detailed heating-rate distributions along complex trajectories. **This capability is currently lacking in the US sensor market, whereas substantial progress has been made by Korean researchers** (9-12). It is critical to characterize unsteady, spatially varying heating loads in high-enthalpy wind-tunnel environments such as Plasmatron X. As part of this task, PI Panerai and Co-I Elliott will work closely with Korean partners to deploy these sensors at Plasmatron X, offering unique testing opportunity, compare them with existing copper slug gauges, and use them to quantify surface catalytic heating.

2) Task 2. Process SiC/SiC materials

Task 2 focuses on the **tailored processing of SiC/SiC ceramic matrix composites** to produce controlled and well-characterized materials set for the aerothermal experiments in Tasks 4. We seek to procure samples made via liquid silicon infiltration (LSI) as baseline processing route, owing to its capability to yield dense SiC matrices, compatibility with woven fiber architectures, and flexibility to tune microstructural features relevant to high-temperature performance. A key emphasis of Task 2 is the **collaboration with Korean industrial partners**, who possess advanced capabilities in SiC/SiC composite manufacturing and high-temperature ceramic processing (13). We seek to leverage their expertise in preform fabrication, controlled carbon deposition, and precision LSI processing to produce high-quality, reproducible specimens for this project. A central goal of this task is to create **material specimens with systematically varied levels of porosity**, enabling direct investigation of how pore volume, pore connectivity, and microstructural heterogeneity affect oxidation kinetics, active-to-passive oxidation transitions, and catalytic heating under hypersonic conditions.

3) Task 3. Develop diagnostics for SiC/SiC volatilization

This task will be dedicated to developing diagnostics to quantify gas-surface interaction mechanisms in SiC/SiC CMCs under hypersonic-relevant conditions, by probing surface-adjacent reactive species, in situ, during Plasmatron X testing. A central focus is targeting passive-to-active oxidation transition, which remains a grand challenge in thermal protection for hypersonics. In passive oxidation, SiC forms a protective SiO_2 layer, limiting mass loss and suppressing volatile product formation. Under high temperature and low-pressure conditions such as high-altitude hypersonic flight, this layer can volatilize or fail, leading to active oxidation, characterized by formation and ejection of SiO, Si, and CO. The onset of active oxidation drives accelerated recession, alters catalytic heating via O-atom recombination on fresh reactive surfaces, and fundamentally changes TPS performance. Resolving salient species from this process, including CO, Si, and SiO, in the first 100-200 μm above the surface is critical to determine reaction pathways, reaction rates, and boundary conditions for coupled aerothermal-chemistry models.

Optical emission spectroscopy (OES) is a low-cost method for identification of active oxidation products, and the method was recently deployed by Co-I Kearney and collaborators (14) to identify SiO and Si atom in ICP testing of SiC-based materials. OES is a good indicator of active oxidation, but this path-integrated technique generally lacks spatial resolution required for gas-surface chemistry model development. The CHESS team will additionally field laser diagnostics for the required model development data at high-spatial resolution. Laser-induced fluorescence (LIF) can provide two-dimensional maps of SiO and Si atom distributions within 100 μm or less of the ablating CMC surface (14). Coherent anti-

Stokes Raman scattering (CARS) provides spatially resolved profiles of the CO product and measurement of the surface-adjacent temperature profile which defines the thermal environment (15, 16). These gas-surface diagnostic methods are combined with catalytic heat flux sensors developed in Task 1, to deliver a complete picture of CMC passive-to-active transition and material performance in the Plasmatron X environment.

4) Task 4. Perform Plasmatron X wind tunnel experiments

In Task 4 we propose to execute a Plasmatron X test campaign that integrates advanced heat-flux sensors, diagnostics and SiC/SiC specimens of controlled porosity to quantify passive-to-active oxidation transitions and catalytic heating effects for CMC materials. We will test at targeted operating points from passive-oxide formation to active oxidation. T**he test campaign will offer the unique opportunity to validate Korean fast-response gauges, asses the performance of Korean manufactured materials and achieve a detailed aerothermal model for SiC/SiC response to hypersonic conditions.** In addition to standard infrared pyrometry for surface temperature measurements, OES/LIF/CARS diagnostics from Task 3 will provide quantification of reactive species and flow temperatures in the reactive boundary layer to identify active oxidation onset and quantify catalytic recombination. By coupons across a designed set of porosity levels we will isolate effects of microstructure controls on oxidation kinetics, catalytic heating and recession.

Data products will consist of a curated aerothermal database including facility conditions species/temperature fields, mass-change and recession metrics, along with associated uncertainty bounds. The resulting database will anchor gas–surface chemistry models and provide traceable, sensor-validated benchmarks for TPS design and future down-selection of CMC materials.

Proposed project timeline
Project Kick-Off (KO): July 2026
Phase I
Preliminary gauges development and environment characterization: 6 months
Milestone 1 (Dec 2026): Heat flux gauges demo
Phase II
Task 1: 6 months
Milestone 2 (Jul 2027): Environmental characterization
Task 2: 6 months
Milestone 3 (Dec 2027): Processed SiC/SiC samples for testing
Task 3: 6 months
Milestone 4 (Jul 2028): OES and LIF demonstration
Task 4: 12 months
Milestone 4 (Jul 2029): Full aerothermal database

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	PI Panerai: lead for Task 4, project management PI Kearney: lead for Task 3 PI Elliott: lead for Task 1 The CHESS team will provide access to Plasmatron X facility, develop characterization and conduct experimental campaigns at Plasmatron X, perform the bulk work for tasked 1, 3, 4, coordinate collaboration with Korean partners.
Expected R&R of Korean SME Partner	Korean Partner 1: • develop heat flux gauges for high enthalpy environment • develop calibration strategy for heat flux gauges • visit Illinois for experimentation • perform data analysis • support meetings and dissemination of project results Korean Partner 2: • develop SiC/SiC composites of controlled porosity via liquid silicon infiltration • characterize properties of composites • deliver samples to Illinois • visit Illinois for experimentation • perform data analysis • support meetings and dissemination of project results

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	There is an urgent need to improve measurement and material technologies to enable hypersonic flight. With the proliferation of high enthalpy facilities in the US to respond to national priority needs in hypersonics, validated sensors that allow accurate characterization of the ground testing environment will be critical and will have great potential for commercialization. There is simply very limited expertise and manufacturing capacity in this area. Similarly, SiC/SiC composite technology needs improvements in high throughput and tailored manufacturing strategies, with high potential for commercialization to the US defense sector.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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